QUICKORDER: ONLINE FOOD ORDERING AND DELIVERY

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1.0 Abstract

In response to the challenges facing the Zimbabwe's food industry where users have to go through generic menus to order their preferred food items and also wait longer times for their deliveries to arrive, I propose this system called quickOrder, which is an AI-powered food ordering and delivery system that utilizes advanced machine learning algorithms to offer personalized food recommendations, optimize delivery routes and enhance overall customer satisfaction. By analyzing user preferences, past order history and user behavior, quickOrder will provide tailored suggestions for food items, alleviating the need for manual menu browsing and enhance engagement. Furthermore, the system incorporates real-time route optimization to ensure accurate delivery estimations and efficient resource allocation, addressing issues of longer waiting times and delayed deliveries. With features like user profiles for storing user preferences and real time order tracking via Google Maps, quickOrder aims to revolutionize the Zimbabwean food delivery sector by providing a seamless, personalized, and efficient ordering and delivery experience to both customers and restaurants.

2.0. Introduction

QuickOrder is an AI powered food ordering and delivery system integrated with machine learning algorithms to analyse customer preferences, optimize delivery routes and streamline operations. By incorporating AI in this system, the system offers personalized recommendations, improve order accuracy and enhance customer satisfaction by suggesting food items and restaurants based on user preferences, order history, ratings and other relevant data.

More so, the system allows users to create profiles where they can save their preferences, dietary restrictions, delivery addresses, and payment information, and hence by storing user preferences, the system can offer more accurate and relevant recommendations, as well as streamline ordering process. Also, by analysing user behaviour and preferences, the system can offer tailored recommendations that match individual tastes and increase the chances of user satisfaction.

Furthermore, to ensure efficient and timely deliveries, the system incorporates delivery route optimization. It takes into account the location to determine the most optimized route for the delivery driver, and to achieving this, factors like distance, traffic conditions, and delivery time windows are considered to minimize delivery time and costs. QuickOrder takes advantages of these machine learning algorithms to enhance customer experience and also increases customer satisfaction and loyalty.

3.0. Problem definition

Some of the primary concerns that are plaguing Zimbabwe Food delivery industry includes lack of personalization and recommendation as customers often has to manually browse through menus to find items of interest, as all the food ordering and delivery system we currently have in Zimbabwe rely on generic menus and do not take into account individual preferences, dietary restrictions, or past order history which can be overwhelming and time consuming leading to customer unsatisfaction and non-engaging experience for customers, therefore restaurants sometimes miss out on opportunities to upsell and provide personalized recommendations using customer preferences. Another problem affecting the Zimbabwean food delivery sector is inaccurate delivery estimations and route optimization as most of the delivery system present in Zimbabwe lacks the capability to optimize delivery routes for drivers effectively, hence this inefficiency longer waiting times for customers, delayed deliveries, dissatisfied customer and inefficient resource allocation for restaurant.

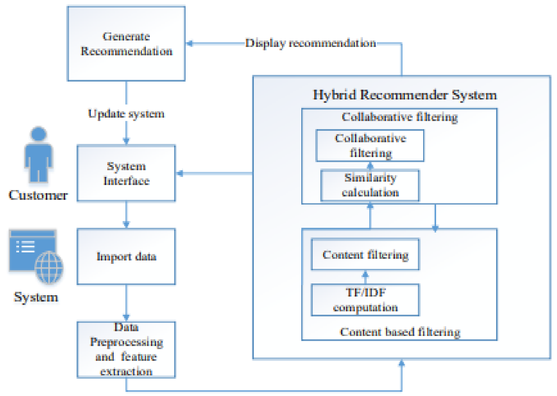
4.0. Overview of the developed application

QuickOrder is a food ordering and delivery system that will be integrated with machine learning algorithms to analyze customer preferences, optimize delivery routes and streamline operations. By incorporating AI in this system, the system offers personalized recommendations, improve order accuracy and enhance customer satisfaction by suggesting food items and restaurants based on user preferences, order history, ratings and other relevant data. Furthermore, to ensure efficient and timely deliveries, the system incorporates delivery route optimization. It takes into account the location to determine the most optimized route for the delivery driver, and to achieving this, factors like distance, traffic conditions, and delivery time windows are considered to minimize delivery time and costs.

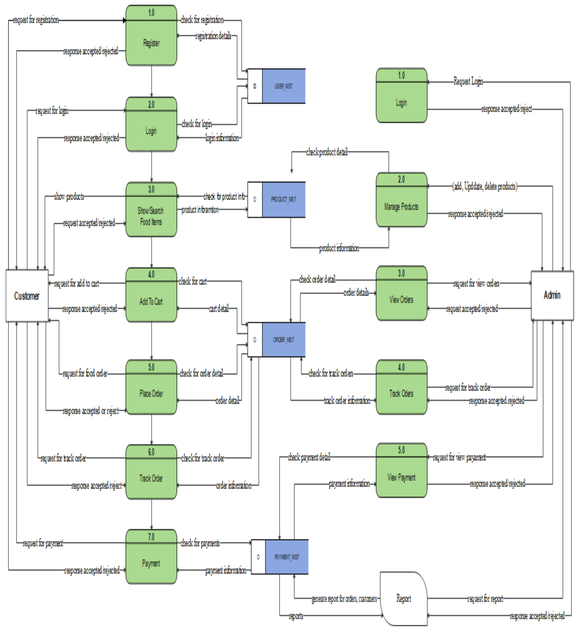
5.0. Objectives

- To provide customers with personalized recommendations of dishes and restaurants they may like by analyzing their preferences, past order history, behavior and individual taste.
- To optimize delivery routes by estimating accurate delivery times and allocating delivery resources efficiently.
- To provide real-time tracking and management of available ingredients, preventing customers from ordering out-of-stock items.
- To provide customers with real time order tracking through the use of Google maps

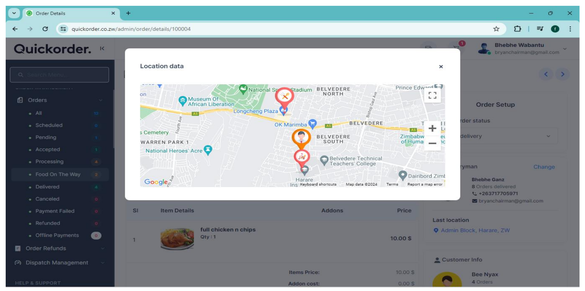
6.0. Recommender Overview

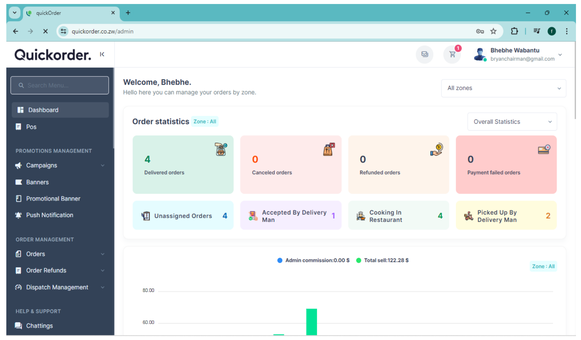


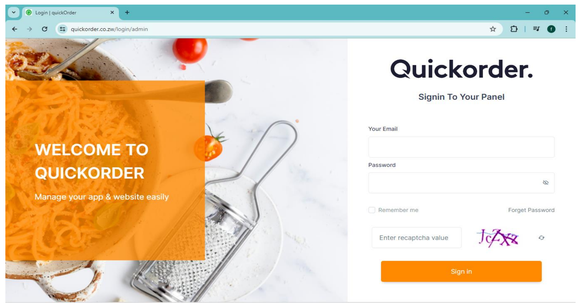
7.0. DFD Diagram



8.0. Screenshots







Conclusion

The integration of personalized recommendations and delivery route optimization can further enhance the user experience in food ordering and delivery systems. By considering both user preferences and delivery constraints, these systems can provide recommendations that not only align with the user's tastes but also optimize the delivery routes for efficient and timely deliveries. More so, this literature review highlighted the various techniques, approaches, the existing research and developments in the area of food ordering and delivery systems with personalised recommendations and delivery route optimisation. The studies provide valuable insight into the design, development, and evaluation of effective systems that can provide information on the development of food ordering and delivery systems with personalised recommendations and delivery route optimisation

Supervisor – Ms Chibaya